

ORACLE RISK

Oracles are one of the last unsolved problems in DeFi and are required by most DeFi protocols to function correctly. Fundamentally, oracles aim to answer the simple question: How can off-chain data be securely reported on chain? Without oracles, blockchains are completely self-encapsulated and have no knowledge of the outside world other than the transactions added to the native blockchain. Many DeFi protocols require access to secure, tamper-resistant asset prices to ensure that routine actions such as liquidations and prediction market resolutions function correctly. Protocol reliance on these data feeds introduces *oracle risk*.

Oracles represent significant risks to the systems they help support. If an oracle's *cost of corruption* is ever less than an attacker's potential *profit from corruption*, the oracle is extremely vulnerable to attack.

To date, three types of oracle solutions have been introduced, developed, and used. The first is a *Schelling-point oracle*, which relies on the owners of a fixed-supply token to vote on the outcome of an event or report the price of an asset. Examples of this type of oracle include Augur and UMA.¹² While Schelling-point oracles preserve the decentralization components of protocols that rely on them, they suffer from slow times to resolution. Second is an *API oracle* a centralized entity that responds asynchronously to requests for data or prices. Examples include Provable, Oraclize, and Chainlink.¹³ All systems relying on API-based oracles must